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To cite this article: Juebei Chen, Anette Kolmos & Xiangyun Du (2021) Forms of implementation and challenges of PBL in engineering education: a review of literature, *European Journal of Engineering Education*, 46:1, 90-115, DOI: [10.1080/03043797.2020.1718615](https://doi.org/10.1080/03043797.2020.1718615)

To link to this article: <https://doi.org/10.1080/03043797.2020.1718615>



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Published online: 09 Feb 2020.



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Forms of implementation and challenges of PBL in engineering education: a review of literature

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ABSTRACT

During the last 40 years, problem- and project-based learning (PBL) has been widely adopted in engineering education because of its expected effectiveness in developing students' professional knowledge and transferable skills. With a growing number of PBL researches and practices in engineering education, systematic or meta-analysis reviews were conducted regarding the definitions, history and development of PBL, and benefits for student learning outcomes. However, challenges in PBL implementation was little addressed in the current review works, and even less attention has been paid on how these challenges in implementation are related to the diverse PBL practices. This paper reviewed 108 research articles to explore the levels at which the currently reported PBL practice is being implemented, and what challenges in PBL practices are being addressed. This research illustrates the variety of PBL implementation at the course level, cross-course level, curriculum level, and project level. Across these four levels, similar challenges are reported at the individual level for teachers and students, as well as at the institutional level and the culture level. Recommendations on future research directions for engineering educational researchers and suggestions for engineering faculty and staff are proposed to optimise PBL curriculum design and inform future PBL implementation.

ARTICLE HISTORY

Received 24 October 2019
Accepted 15 January 2020

KEYWORDS

PBL implementation;
engineering education;
challenges

1. Introduction

The main responsibility of practising engineers lies in solving uncertain, complex, open-ended work-place problems (Jonassen, Strobel, and Lee 2006). With these demands, problem- and project-based learning (PBL) has been widely adopted in engineering education during the last 50 years because of its expected benefits in improving students' academic achievement and transferable skills (Kolmos and Fink 2004). PBL implementation strategies in different culture settings and educational systems range from the curriculum level to the course level, from being problem-orientated to project-orientated, and from being one subject to multidisciplinary, but the fundamental principle remains that students are motivated by themselves and become owners of the learning process (Edström and Kolmos 2014; Lehmann et al. 2008). With various PBL implementations, over the decades several systematic or meta-analysis reviews have provided a summary of evidence on the history of PBL, diverse PBL designs, students' learning outcomes, and its effectiveness in higher education (Dochy et al. 2003; Thomas 1997). Several meta-analysis reviews have been conducted by educational researchers, including Dochy et al. (2003), Strobel and Barneveld, and Gijbels et al. (2005),

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aimed at identifying the definitions of PBL, discussing the theoretical foundation of PBL, and addressing the effectiveness of PBL in students' knowledge and skills development. Most of the reviews are within PBL and medicine, which do practise case-based PBL, whereas PBL in engineering education is mostly project-based. In engineering education, research on PBL practice has also boomed in recent years, mainly focusing on the illustration of PBL design, the effectiveness of PBL, the measurement of students' learning outcomes in PBL, and comparing between PBL and non-PBL students (Beddoes, Jesiek, and Borrego 2010; Johnson and Ulseth 2016; Perrenet, Bouhuijs, and Smits 2000). Perrenet, Bouhuijs, and Smits (2000) reported on several PBL projects at Maastricht University and Aalborg University, and pointed out the effectiveness of PBL in students' acquisition of knowledge and skills as well as the possible limitations of PBL as an overall strategy for engineering education. However, the functions of PBL are not sufficient and conclusive, possibly because the current literature is based on an assumption that PBL is uniform, but it neglects the diversity in practice, therefore the issues addressed also vary in the different methods of implementation. More attention needs to tease out and summarise an overview of the current diverse PBL practices.

While an increasing number of engineering projects have implemented PBL at different levels, several studies have reviewed the history of PBL and PBL cases at the theoretical level; however, little literature has been reviewed concerning the implementation levels of PBL design or the current challenges for PBL practice. In addition, previous studies reviewed the development of PBL from the end of the twentieth century to the early twenty-first century (Dochy et al. 2003; Savery 2015). Reis, Barbalho, and Zanette (2017) reported a classification and bibliometric analysis of PBL research in engineering, which focused on research design instead of the types of PBL practice. With a growing number of studies having reported their PBL practices in recent decades, an updated review is needed to create a broader overview of the current PBL implementation. Thus, this literature review aims to map the current PBL practice in different institutions and expand our understanding of challenges reported in empirical PBL research from engineering education. Based on PBL principles addressed in previous research, PBL cases where students have become the centre of learning and work as teams to solve problems/projects in the field of engineering education (Du 2006; Edström and Kolmos 2014) are gathered and reviewed to scrutinise evidence of current PBL implementation with relevant empirical research to answer the following questions:

- 1) At what levels is the currently reported PBL practice being implemented in engineering education?
- 2) What kinds of challenges in PBL practices faced by both students and teachers are addressed in current studies in the field of engineering education?

In order to further identify the types of reported PBL practices and connect the theoretical framework of PBL types with current various PBL implementations, we referred to Savin-Baden's (2014) conceptional framework of constellations of PBL implementation. This framework illustrates nine types of PBL practices: PBL for knowledge management, PBL through activities, project-led PBL, PBL for practical capability, PBL for design-based learning, PBL for critical understanding, PBL for multimodal reasoning, collaborative distributed PBL, and PBL for transformation and social reform. With a mapping of PBL constellations, Savin-Baden pointed out that more attention should be paid to connecting the theoretical framework with current PBL design, with the aim of clarifying the purpose, practices, and pedagogies involved in PBL implementation. Based on a literature review and analysis, this study maps diverse PBL practices at different levels and current challenges for PBL implementation. Among multiple types of PBL practices, teachers have played different roles in the PBL process. Many researchers have used 'instructor' in their articles when teachers had more instructions for students about what to do. In other articles, 'facilitator' has been used since teachers' responsibility was to facilitate students' PBL process. In this article, we use 'teachers' in general to include those two types of teacher roles. This article summarises current PBL practices and challenges reported in literature from 2000 to 2019. Based on the overview of PBL implementation provided by this article,

future research directions are identified for engineering education researchers. Suggestions are also proposed for engineering faculty and staff to optimise their PBL curriculum design and inform PBL implementation.

2. Methodology

Instead of meta-analysis which evaluates the qualities of papers, our research objectives focus on providing the spectrum of the divers current PBL implementation at different levels. Thus, the systematic review method is adopted to enable researchers to identify current reported PBL practices as much as possible. In order to synthesise current literature, a research review typically includes the following steps (Borrego, Foster, and Froyd 2014; Petticrew and Roberts 2008): (1) identify the problem with preliminary scoping; (2) formulate specific research questions; (3) design the research protocol with inclusion criteria; (4) analyse and synthesise the data; (5) report the findings with a detailed description of review procedures. With the identified problem and proposed research questions, the research protocol and searching process are conducted following the instructions of the literature review in engineering education proposed by Borrego, Foster, and Froyd (2014).

2.1. Developing the research protocol

Based on research questions and the definition of PBL, criteria were set for inclusion in review and analysis: (1) All selected researches were peer-reviewed empirical work, written in English. Although several literature reviews were also included, they were only used for snowball searching rather than being analysed; (2) All research should be conducted in the context of PBL and fit our previous description of PBL practice; (3) Studies were conducted in the context of engineering education in colleges or universities. Secondary school education is excluded; (4) In order to enhance or maximise ecological validity, selected empirical works should be conducted in a real-life classroom or learning projects instead of under controlled laboratory situations (Gijbels et al. 2005).

A time frame of 2000–2019 is set for the selection of articles for this study in order to provide an updated review of current PBL implementation in the past two decades. Selected from databases recommended by Borrego, Foster, and Froyd (2014), we conducted a literature search in the following four databases: EBSCO (academic search and education full text), ERIC (ProQuest), Web of Science, and SCOPUS. These databases include conference papers, journal articles and book chapters with divers perspectives on practice focus, research focus, national audiences, and international audiences, which enable us to explore the variety of current PBL practices across countries in engineering education. Based on a variety of keywords used in studies on PBL practice in engineering education, a series of keywords with alternative words were set for searching (Table 1).

2.2. Searching and filtering

Following Borrego, Foster, and Froyd’s (2014) instructions on filtering methods, which include steps of searching, screening, and appraising, we conducted a literature search in the four databases with selected keywords in June 2019. As can be seen in Figure 1, the search results included 436 articles in

Table 1. The series of keywords for searching.

Block 1	'PBL' or 'problem-based' or 'project-based' or 'problem solv*' or 'problem-orient*' or 'project-orient*'
	AND
Block 2	'engineering education' or 'manufacturing'
	AND
Block 3	'implement*' or 'practice*' or 'practise*'
	AND
Block 4	'challenge*' or 'issue*' or 'difficult*'

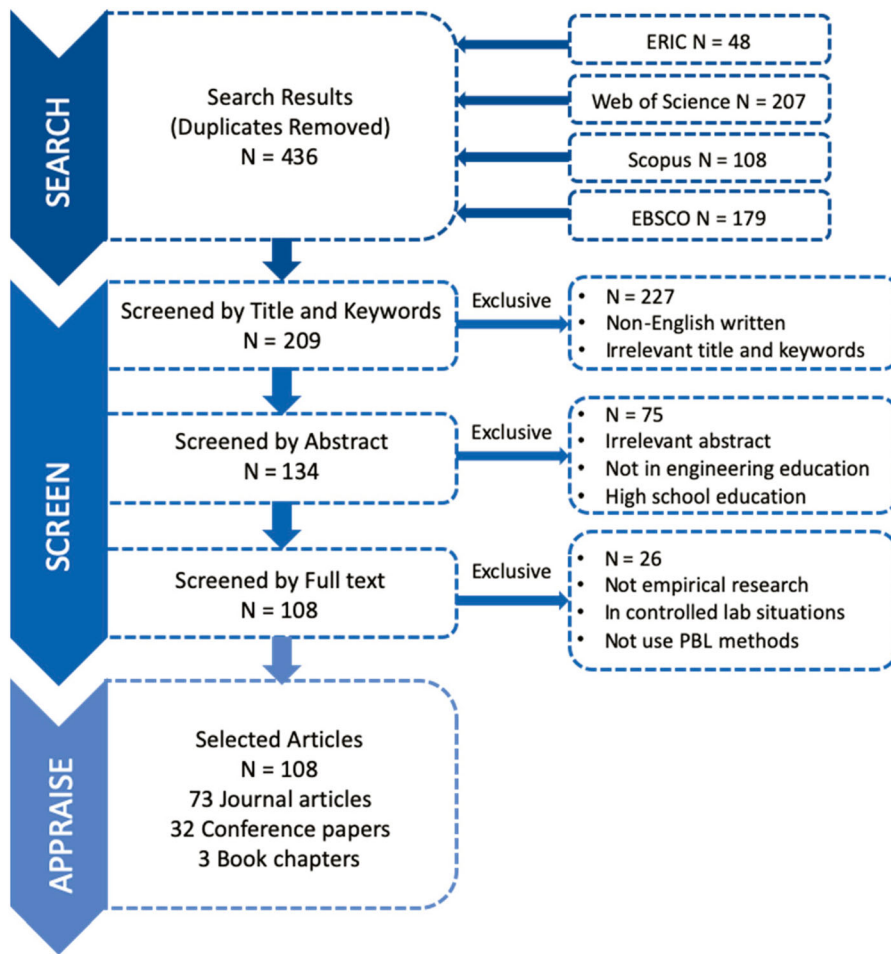


Figure 1. Flow chart for searching and filtering process.

total. Articles were first filtered based on the relevance of their titles and keywords, and then a second round of filtering was conducted via abstract analysis. Seventy-five articles were removed due to irrelevant content, which focus of high school education or not engineering education. In the next step, full-text reading was conducted among the remaining articles to ensure that the selected articles met our inclusion criteria. Twenty-six articles were excluded because of several reasons, such as they are not empirical research, they did not follow the characteristic of PBL, or they were in the situations of controlled laboratory. The final number of selected articles was 108, including 73 journal articles, 32 conference papers, and three book chapters.

2.3. Demographics of selected publications

In order to map current PBL implementation in engineering education, demographic information was classified into the following aspects: country, publication source, subjects, and research methods.

The majority of researchers are American, representing 27 of the 108 selected articles. Authors also come from Spain (16), China (10, including Hong Kong), Australia (8), Denmark (7), the United Kingdom (6), Brazil (5), Malaysia (4), and France (3). Two articles each are from the United Arab

Emirates, Canada, Ireland, Korea, Turkey, and Ukraine, and there is one each from Georgia, India, Indonesia, Israel, Japan, the Netherlands, Palestine, Portugal, Romania, and Sweden.

In terms of publication source, there are 73 journal articles, including from the *European Journal of Engineering Education* (17), *IEEE Transactions on Education* (12), the *International Journal of Engineering Education* (3), the *Journal of Problem-based Learning in Higher Education* (3), *Innovations in Education and Teaching International* (2), the *International Journal of Sustainability in Higher Education* (2), the *International Journal of Technology and Design Education* (2), *Advances in Engineering Education* (1), the *American Journal of Engineering Education* (1), *Higher Education Pedagogies* (1), and the *Interdisciplinary Journal of Problem-based Learning* (1) etc. Thirty-two conference papers are included in this study, most of which are from the ASEE Annual conference (14), followed by IEEE Frontiers in Education Conference (3), the World Engineering Education Forum (2), the IEEE Global Engineering Education Conference (1), the SEFI Annual Conference (1), and other international conferences in specific engineering fields.

The subjects of selected articles that presented their PBL mainly focus on the field of computer and software engineering (27), followed by electronics engineering (16), mechanical engineering (13), and civil engineering (11). Other researches were conducted in the fields of biomechanical engineering, chemical engineering, geotechnical engineering, energy engineering, and so on. Six articles reported their multidisciplinary PBL practice, which includes students from diverse engineering subjects as well as students from business and social science.

Qualitative studies comprised the majority of research types among all selected articles. Among all qualitative studies, 46 studies adopted case study design to illustrate their PBL design, students' cases, experience and feedback in detail. Other 14 qualitative studies utilised interviews, observation, and qualitative surveys as research methods. There are 45 studies using quantitative methods, including surveys and questionnaires. Two articles utilised mixed methods, combining observation or interviews with surveys. One article used action research design.

2.4. Tracking and analysis

Based on the objective of providing an overview of divers PBL implementation, initial themes were set up through group discussion with three researchers, to guide the coding of selected articles in the analysis processes, including levels of PBL implementation, types of problems for students to solve, project duration, group size, assessment methods, and challenges faced by students and teachers. In order to develop a structured codebook, initial coding was done by the lead coder for ten articles, random selected among all 108 articles. Based on initial themes, the open-coding method was adopted, and codes were developed, discussed, categorised, and summarised into subthemes under the initial themes (Creswell 2002). Both subthemes and initial themes comprise the focus of the findings presented in this literature review. In addition, in order to explore the various types of PBL practices, Savin-Baden's (2014) theoretical framework of types of PBL design served as a reference in this process, and includes PBL for knowledge management, PBL through activities, project-led PBL, PBL for practical capability, PBL for design-based learning, PBL for critical understanding, PBL for multimodal reasoning, collaborative distributed PBL, and PBL for transformation and social reform. With the guidance of Savin-Baden's (2014) definitions of PBL types, descriptions of learning emphasis, activity design, and assessment methods in selected articles were coded and analysed in this process for categorisation when authors provided enough related information.

In order to improve the validity of data analysis, selected papers were read multiple times and coded by the lead coder with researchers' self-reflection. To minimise the researcher's bias, the researcher triangulation with one PhD student researcher and two experienced researchers were conducted as auditing procedures. In the initial coding process, initial themes and codes were modified and refined for two rounds through group discussion. In order to enhance the inter-coder reliability, one graduate student researcher who was experienced with qualitative research but not familiar with this research was also invited to code eight articles using the initial codebook

independently. Resulting in agreement over 88%, codes are modified and/or refined during and after the auditing process through discussing the results of the coding. In addition, after main coding process, a sample of 30 papers and the list of codes and their definitions were provided to a second researcher with rich qualitative research experience. The sample was selected intentionally and cover most codes at all PBL implementation levels. An auditing procedure was conducted to discuss the results of the coding, levels of PBL implementation, and types of PBL. The categories were modified and/or refined in the research groups during and after the auditing process.

3. Findings

3.1. PBL implementation at different levels

This study reviewed 108 articles about PBL implementation from 2000 to 2019 at the course level ($N = 73$), cross-course level ($N = 6$), curriculum level ($N = 23$), and project level ($N = 6$) (Table 2; Appendix, Table A1). At the course level, PBL methods, used as teaching tools, were added into lectures or workshops in order to train students in self-learning skills and arouse students' enthusiasm for solving problems (Gratchev and Jeng 2018). This type of PBL practice is adopted in one course and usually lasts for one semester. PBL at the cross-course level means that a series of related or multidisciplinary courses were combined together to support students' project in a period of time, mostly in one semester or two semesters in several cases (MacCrum 2017). At the curriculum level, PBL constitutes the backbone of a curriculum while other traditional learning methods such as lectures become assistant elements to support PBL (Perrenet, Bouhuijs, and Smits 2000). At this level, PBL is used as the main learning method in a large part of or throughout the whole professional training. PBL at the project level means that short-term (one or two months) or long-term (six months to one year) project (such as summer schools and joint educational project) using PBL methods are organised by one university, joint higher education institutions, or universities with companies. In contrast to PBL at the curriculum level, PBL project at this level are mostly outside the professional curriculum or co-curriculum and do not require all students to participate in them, where students have the freedom to participate or the project are not based on their own interests.

In this article, PBL contains both problem-based learning and project-based learning. At the course level, both problem-oriented and project-oriented learning methods were adopted by engineering educators, but at the cross-course, curriculum and project levels, project-oriented PBL were more used, where students were required to work on real-world problems/design/products as group projects. In addition, based on the overview of current PBL practice, PBL at the course level is the most widely used implementation among all selected articles, following by PBL at curriculum level, which is reported by researchers from Denmark, Spain, Ireland and the U.S. There are six cases of cross-course

Table 2. An overview of PBL practice levels.

Level	Number	Duration	Implementation	Organizers	Discipline
Course	73	Mostly one semester	Inside the professional training design	Teachers or departments in universities	Mostly one discipline
Cross-course	6	Mostly one semester	Inside the professional training design	Departments in universities	Mostly multidisciplinary
Curriculum	23	Varies from 2 years to 4 years	Inside the professional training design	University	A combination of one-discipline and multidisciplinary projects/problems
Project	6	Varies from short-term to long-term	Outside the professional training design	One university or joint universities or universities with companies	Multidisciplinary

PBL practices from the UK, Spain, United Arab Emirates, the U.S., and Australia. Six cases of short-term or long-term PBL projects are reported by researchers from China, Denmark, Japan, Thailand, Spain, and the U.S.

3.1.1. PBL practices at the course level

PBL practice at the course level, where PBL elements were added into one existing course, are reported with the highest frequency among all selected articles ($N = 73$). Based on Savin-Baden's (2014) framework of PBL, several modes of implementing PBL were included at this level (Table 3), including PBL for knowledge management, PBL through activity, project-led PBL, PBL for practical capabilities, and PBL for design-based learning. We did not count the frequency of each PBL type at the course level since many courses combined several types of PBL design in their practice, but examples of each PBL type are shown below.

In the practice of PBL for knowledge management, which aims to test students' knowledge acquisition, students were expected to discuss and solve in class well-structured/semi-structured problems proposed by teachers, and their learning outcomes of professional knowledge were tested through several traditional assessment methods, such as attendance, quizzes, homework assessment, and midterm and final exams (Dong and Guo 2014; Hu, Ortiz, and Sriraman 2014). In one PBL practice case, PBL units were set following lecture and lab experiments in order to further enhance students' understanding and use of knowledge (Williams 2011). In another case, Requies et al. (2018) reported that their students needed to conduct PBL (first 8 weeks) at the beginning of the course in order to let them experience the real processes for working as engineers first. Among PBL practice reported in selected articles, project-led PBL is widely used at course level. In those cases, students were required to solve one or more real-world, open-ended, and ill-structured projects in teams (with 3–8 members) in one semester. Students' learning outcomes were assessed using team project reports, team oral presentations, individual oral reports, and peer assessment (Costello 2017; Garcia-Robles et al. 2009; Hassan et al. 2015). In the practice of PBL for practical capabilities, students formed small groups to solve practical and multidisciplinary problems in one semester, with the intention of developing students' multidisciplinary knowledge and competence for real-world problems (Chaparro-

Table 3. Types of PBL practice at the course level.

	PBL for knowledge management	PBL through activity	Project-led PBL	PBL for practical capabilities	PBL for design-based learning
Problem types	- Semi-structured problems - Proposed by teachers	- Authentic and open-ended problems - Proposed by students	- Real-world, open-ended, and ill-structured projects - Identified by students or given by teachers	- Practical, real-world, and multidisciplinary problems/cases - Identified by students or given by teachers	- Real-life design - Proposed by students or given topics from teachers
Duration	In one class	One or two learning modules of the course	One semester one or more projects	One semester	One semester
Team size	Not mentioned	4–5 students	3–5 students	Small groups	3–8 students
Assessment methods	- Homework assessment - Individual quiz - Midterm exam - Final exam - Attendance	- Course paper - Presentation - Lab reports	- Team written reports - Team oral presentation - In-group peer assessment - Individual oral presentation ...	- Individual exercise - Individual final reports - Team reports and presentation - Peer assessment ...	- Team report and presentation - Peer review - Self-assessment - Individual contributions in teams - Final exams - Attendance ...

Table 4. Types of PBL practice at the cross-course level.

Authors	Type of PBL	Learning emphasis	Problem types	Duration and arrangement	Team size	Assessment methods
McCrum	PBL for critical understanding/ PBL for design-based learning	- Synthesis across boundaries - Design-based capabilities	- Complex, open-ended problems - Real-world design - Proposed by students	11 weeks - Supported by interdisciplinary learning modules	Not mentioned	- Presentation - Structural drawings - Project report
Calvo et al.	PBL for critical understanding	- Synthesis across boundaries - Design-based capabilities	- Real-world product design - Provided by teachers	15 weeks - Supported by interdisciplinary courses	Three students	- Final team presentation - Individual programming exams - Final exam
Chowdhury	PBL for practical capabilities	Development of capabilities	- Complex and authentic problems - Given by teachers	- One semester - Supported by six multidisciplinary sessions	Three students	- Peer review - Self-evaluation
Wallace and Webb	PBL for practical capabilities/ PBL for design-based learning	- Communication skills - Problem-solving skills - Design skills	- Real-world problems in STEM - Public product design	- One semester - Supported by professional courses - Courses about communication skills	Not mentioned	- Final workshop product - Team oral presentation - Team written reports
Kay et al.	PBL for practical capabilities	Development of capabilities	- Open-ended, authentic, substantial problems - Given by teachers	4 weeks for pilot study 9 weeks for main projects - Supported by professional courses	Small group	- First problem: individual assessment - Second problem: group report; presentation; reflective report; weekly plan; individual exam
Chen and Teng	Collaborative distributed PBL	Collaborative learning and effective teamwork	- Real-life problems at different levels of difficulty - In software engineering - Given by teachers	- Two semesters - Meetings with stakeholders - Supported by professional courses	5–6 students - One instructor - One sponsor from company	- Project grading - Attendance and engagement report - Stakeholders' observation - Satisfaction survey and interview

Peláez et al. 2013; Hartman and Gindy 2010). In the type of PBL for design-based learning, students were required to design real-life products in teams, which usually last for one semester. In several cases, the real-life designs were combined with students' capstone projects (Gweon et al. 2017). Not only innovative assessment methods such as peer review and self-assessment but also traditional assessment (final exam and attendance etc.) were adopted in reported PBL practice (Huang and Yang 2008; Lutsenko 2018; Valdez et al. 2009).

3.1.2. PBL practices at the cross-course level

At the course level, there were six articles introducing their PBL implementation, three of which had a set of multidisciplinary engineering courses (Calvo et al. 2018; MacCrum 2017) while the other three articles reported their PBL cross-course practice in one engineering subject (Chen and Teng 2011; Kay et al. 2000; Wallace and Webb 2017). According to descriptions of learning objectives and PBL design in each article, we also categorised their PBL practice at the cross-course level into Savin-Baden's (2014) framework (Table 4). In all six articles, students are required to solve real-world problems/design, but with various arrangements of learning activities, their PBL practice had a different learning emphasis. For example, as reported by Calvo et al. (2018), their PBL implementation was supported by multidisciplinary courses, aimed at training students' transferable abilities across contexts. In another case of collaborative distributed PBL practice, Chen and Teng (2011) emphasised the importance of collaborative skills and providing the chance for students to solve real-life problems and communicate with stakeholders from companies. Students were assessed by both teachers and stakeholders from companies through group meetings, individual performance, and project reports. Compared with PBL practices at the course level, growing complexity could be seen at this level. At the cross-course level, students had more chances to solve complex, open-ended, multidisciplinary, and real-world problems instead of well-structured problems that were used to test knowledge. PBL practices were set with various learning objectives focusing on transferable skills, such as multidisciplinary knowledge and skills, critical thinking skills, and teamwork skills, and thus we found more practice types of PBL for critical understanding, PBL for practical capabilities, and collaborative distributed PBL at the cross-curriculum level.

3.1.3. PBL practices at the curriculum level

PBL at curriculum level refers to the use of PBL methods in curriculum design in order to achieve specific learning outcomes. At this level, PBL is not limited to one course but used as one of the key learning methods in undergraduate or graduate programmes in one higher education institution ($N = 23$). Several universities stated that they have implemented PBL methods at curriculum level, including Aalborg University (Balve, Krüger, and Tolstrup Sørensen 2017; Dahms, Spliid, and Nielsen 2017; Guerra 2017; Hosseinzadeh and Hesamzadeh 2012; Zhou 2012), University College Dublin (Ahern 2010), the Universidad Europea de Madrid (Terrón-López et al. 2015, 2017), and Clemson University (Qattawi, Venhovens, and Brooks 2014). This level included three types of articles: (1) eight articles introduced the whole four-year PBL curriculum; (2) eight articles reported their usage of PBL methods as a part of the curriculum (1–3 years); (3) seven articles gave an example of their PBL practice in curriculum systems (1-semester or 2-week projects).

Taking an overview of PBL practice at curriculum level, students were also required to solve real-world, open-ended, and interdisciplinary problems in small groups (2–8 students). These problems might be defined by students (Dobson and Tomkinson 2012; O'Shea et al. 2013; Tan and Shen 2018) or by teachers (Hosseinzadeh and Hesamzadeh 2012; Li 2015; Schultz 2018). One special thing reported at this level is that in several cases, real-world problems for students to solve were proposed by real clients or companies directly, which required students to work as real engineers, listen to clients' needs and communicate with outside stakeholders (Terrón-López et al. 2015, 2017; Wallace and Webb 2017; Zhou 2012). In the processes of assessment, team oral presentation, project report posters, peer evaluation, and weekly group relationship assessment were used to measure their team performance and outcomes (Ahern 2010; Dahms, Spliid, and Nielsen 2017;

Table 5. Types of PBL practice at the project level.

Authors	Type of PBL	Learning emphasis	Joint partners	Problem types	Duration and arrangement	Team size	Assessment methods
Mantelet et al.	PBL for design-based learning	Capabilities of design, creativity and innovation	–	- Real-world design - General topics proposed by teachers - Specific projects proposed by students	- One small project one day - Co-curriculum	8 students	- Design graded by four experts - Group presentation
Mativo et al.	- PBL for practical capabilities - Collaborative distributed PBL	- Development of capabilities as real engineers - Collaboration skills	Industrial partners	- Real-life and open-ended problems - Proposed by industrial partners	- Four weeks for one project, two semesters - In formal curriculum	3–4 students	- Midterm and final project reports - A survey on teamwork
Ota and Punyabukkana	- PBL for multimodal reasoning - PBL for transformation and social reform	- Capabilities to solve global issues - Critical thinking skills	- Tokyo Institute of Technology in Japan - Chulalongkorn University in Thailand	- Practical global issues - Themes proposed by teachers	15 weeks - Co-curriculum	2–3 students	70% individual performance (group contributions, report chapters in charge, and personal essay) 30% group performance (overall report, group presentation, and group discussion)
Moliner, et al.	PBL for practical capabilities	Development of capabilities	Three Spanish universities	- Real-life product design - General topics given by teachers	- One semester - Co-curriculum	Five students	- Individual tests - Individual and team video - Final design reports - Online platform using frequency
Du et al.	- PBL for multimodal reasoning - Collaborative distributed PBL	- Critical thinking skills - Innovation and transferable skills - Collaborative skills cross-context	- Aalborg University (Denmark) - Beijing Normal University (China)	- Real-life and multidisciplinary problems - Topics given by teachers - Specific problems identified by students	- Six months - Co-curriculum	7–9 students	- Final exams - Team reports - Team oral presentation - Reflection essays - Attendance
Sanger and Ziyatdinova	- PBL for multimodal reasoning - PBL for practical capabilities - Collaborative distributed PBL	- Critical thinking skills - Development of capabilities - Collaborative skills cross-context	Industrial partners	- Complex, real-world, and open-ended problems - Given by industrial companies	- One academic year - Co-curriculum	Not mentioned	- Peer review - Grades from mentors - Global exhibition - Performance assessment ...

Hung 2011; O'Shea et al. 2013; Palmer and Hall 2011; Qattawi, Venhovens, and Brooks 2014; Tan and Shen 2018; Wallace and Webb 2017). In the dimension of individual assessment, individual project reports and other traditional assessment methods, such as attendance, essays and exams, were also utilised to measure students' academic achievement (Qattawi, Venhovens, and Brooks 2014; Simcock et al. 2007; Tan and Shen 2018).

At this level, a combination of PBL practice types was set for students in different semesters in order to fulfil specific learning objectives. For example, in Ahern's (2010) article, a two-year PBL curriculum was designed for third-year and fourth-year bachelor students. Although both third-year and fourth-year students were required to identify and solve real-world and open-ended problems, projects for third-year students focused on training students' ability to transfer knowledge into practice (PBL for practical capabilities), while fourth-year projects were aimed at developing students' design-based capabilities (PBL for design-based learning). Due to the multiple types of PBL practices they used in the whole curriculum and the limited cases they reported (one semester, one year, or two years in a four-year study), we did not limit their PBL curriculum design to one specific PBL type.

3.1.4. PBL practices at the project level

PBL practice at the project level means projects using the PBL method are outside the professional curriculum systems, and students have the freedom to participate or not based on their own interests. At this level, PBL cases include short-term (less than 1 month) projects ($N = 2$) and long-term (4 months to 1 year) projects ($N = 4$). Details are shown in Table 5 below.

Among all six cases of PBL practice at the project level, one project was held in an institution where students needed to deal with one real-world design in one day (Mantelet, Segonds, and Jean 2018). Another project was held by three joint Spanish universities, which have built a university network to share education resources and PBL experience in order to improve teaching quality (Simcock et al. 2007). Students with different backgrounds from three universities (Universidad del País Vasco, Universitat Jaume I de Castelló, and Universidad de Cadiz) were enrolled in the project and divided into 12 groups (5 students each) to finish a real-life product design. Project reports, group meeting records, and frequency of platform usage were used to assess students' learning outcomes. Two PBL projects with collaboration between universities and industry were reported at this level. In these cases, students were required to solve problems presented by their industrial partners, which enabled them to develop capabilities as real engineers and communication skills.

In addition to local collaboration, international PBL projects were also reported in selected articles. For example, in China, similar international projects have also been developed by Chinese universities. In collaboration with Aalborg University (Denmark), Beijing Normal University established a six-month PBL-based multidisciplinary projects (Du, Su, and Liu 2013). In Japan, Tokyo Institute of Technology developed a 15-week joint PBL project with Chulalongkorn University in Thailand (Ota and Punyabukkana 2016). In order to develop students' international horizon and critical thinking skills, students in this project solved global and practical problems in two- or three-member teams through collaborative contextual learning, and they were assessed based on their performance in project reports, team presentations, and collaboration reports.

3.2. Current challenges in PBL practice

At each level, we found a huge variation in PBL practices, and when analysing the challenges, it was found that many of the challenges were not connected to a special practice but were found to be general challenges across the four levels. Similar challenges faced by students, teachers, and institutional managers were reported at different PBL practice levels. Thus, instead of the four levels in PBL practice, we categorised challenges reported with high frequency into other three levels: individual level, institutional level, and culture level (Appendix, Table A2). Challenges at the individual level means issues or difficulties faced by individuals, such as students and teachers, in their PBL courses/projects/programmes. Institutional level refers to the perspective from engineering faculty,

departments and universities. Culture level includes components related to culture background, such as languages, nationalities, and cultural custom.

3.2.1. Individual level

Lack of facilitation training for teachers: As reported in many articles, teachers were facing the challenge of transferring their roles from lecturers to facilitators in the PBL environment (Chan 2016; Lutsenko 2018). Without theoretical knowledge and skills regarding PBL methods, in many cases teachers met difficulties in how to design the course activities, how to facilitate students' teamwork, how to balance between helping and influencing the students' work, and so on (Bani-Hani et al. 2018; Santos-Martin et al. 2012; Zhou, Nielsen, and Kolmos 2011). In order to improve PBL implementation and effectiveness, it's important to provide PBL pedagogical training for engineering staff to learn PBL pedagogy knowledge and facilitation skills.

Challenges for choosing effective assessment methods: When designing the PBL courses or programmes, how to choose effective assessment methods is another challenge for teachers. One of the learning objectives for PBL is developing students' transferable skills, such as communication, teamwork, problem solving, and self-directed learning (Edström and Kolmos 2014). However, it's difficult to measure the development of students' transferable skills and actual learning outcomes in one course (Chaparro-Peláez et al. 2013; Kunberger 2013; Macias-Guarasa et al. 2006; Marti et al. 2015). For the purpose of enhancing the validity of learning outcomes assessment, new assessment methods (self-assessment, peer review, presentations, observation, etc.) combined with traditional assessment methods (quizzes, exams, lab reports, attendance, etc.) were widely used in PBL courses. Nonetheless, how to make students remain honest in the self- and peer assessment process is another challenge for teachers (Ahern 2010; Du, Su, and Liu 2013; Nunez, Caro, and Gonzalez 2016).

The Need of continuous PBL skills training for students: In the PBL environment, not only teachers but also students need to transfer from traditional learning methods to PBL methods. Students' previous teamwork experiences could influence their motivation and performance in PBL. However, not all students have experienced teamwork and PBL, or have positive attitudes about their prior teamwork experiences. For those without rich teamwork and PBL experiences, the lack of PBL skills training led to them meeting difficulties over how to deal with conflicts, how to collaborate with students from diverse subjects, how to deal with free-riders in teams, how to communicate effectively in teams, and so on (Bani-Hani et al. 2018; Dong and Guo 2014; McQuade et al. 2018). In addition, students are expected to be the centre of learning in the PBL context, which requires higher self-learning skills for students. Several researchers pointed out that in the self-learning process, it was challenging for students to identify problems and translate knowledge into practical problem solutions (Ahern 2010; Bledsoe and Flick 2012; Hu, Ortiz, and Sriraman 2014; Lutsenko 2018). For those who lacked learning motivation or self-reflection, the effectiveness of PBL and teamwork could be weakened (Gratchev and Jeng 2018; Williams 2011). One suggestion to improve students' PBL skills is to provide continuous guidance from teachers and facilitators on teamwork and self-learning. When designing the PBL course/curriculum/project, the part of pedagogical and transferable skills training should be added, which could be the forms of lecture, workshop, seminar, pilot project, regular meeting and other learning activities to introduce teamwork theories, communication skills, time and team management tools etc.

3.2.2. Institutional level

More time and effort devotion: Current PBL practice emphasises developing students' capabilities to solve real-world, open-ended, and ill-structured problems, which has brought new challenges for both teachers and students in devoting more time and effort. For teachers, instead of just giving lectures and setting exams in traditional teaching courses, they faced a heavier workload in PBL courses by providing professional guidance, practical experiences, and teamwork facilitation during the whole process of finishing projects (Dos Santos et al. 2009; Hassan et al. 2015; Palmer and Hall

2011). For students, working as real engineers and dealing with the messiness of problems required them to devote more efforts to overcoming expected and unexpected issues, and some of them even experienced anxiety or depression in this progress (Kay et al. 2000; Servant and Dewar 2015; Zhou 2012). Thus, when designing PBL courses or programmes, considering what workload and strategies are feasible to motivate both teachers and students to devote more time and effort is also challenging for higher institutions.

Lack of support from departments and institutions: At the system level, the biggest challenge reported in selected articles was the lack of support from their departments or institutions (Chan 2016), especially in applying PBL at the course level. Teachers reflected that they need more supportive materials, resources, and policies from the faculty or university level to improve the effectiveness of PBL (Arman 2019; Clyne and Billiar 2016; Hosseinzadeh and Hesamzadeh 2012). Workshops are needed for PBL teachers to share experience with each other and enhance collaboration between different units in a university or across universities (Arman 2019; Kjellberg, Adawi, and Brolin 2015). For students, the lack of facilitators and resources might also be a challenge. In several cases, students pointed out that they did not get enough guidance and help from facilitators or supervisors (Sahin 2010; Setiawan 2019). Institutions are expected to provide more space and freedom for students to explore their own interest in the PBL context since in many curriculum designs, topics are given by teachers (Ribeiro and Mizukami 2004, 2005).

Challenges for effective PBL design: Although many researches have pointed out that PBL could benefit students in terms of the development of competences, it is still a challenge for higher education institutions to design an effective PBL curriculum since there are too many aspects to consider, such as: how to design learning objectives and learning principles; which PBL models could be transferred into their own institutions; and how to balance PBL projects with professional courses in curriculum design (Dos Santos et al. 2009; Kim 2012). For those PBL practices at the course level, researchers are also concerned that PBL in single courses lacks repercussions in the whole curriculum, which might reduce the effectiveness of PBL on students' longitudinal development (Arman 2019; Du et al. 2019). Thus, PBL curriculum designers in colleges and universities also need pedagogical theories and PBL knowledge in order to design a professional curriculum with a combined perspective of learning theories and professional standards. In order to design a structured PBL practice, educational researchers, engineering staff and university managers need the horizon beyond single engineering courses and set the sustainable learning objectives for students' longitudinal development in PBL.

Limitation of external conditions: For those institutions that have used traditional teaching methods for decades, one challenge of the PBL curriculum reform is the limitation of external conditions, including a high student-staff ratio, a lack of infrastructure for teamwork, a lack of technical support, and limited financial support (Clyne and Billiar 2016; Huang and Yang 2008; Ruhizan, Ramlee, and Azami 2009). Although the aim of PBL is to enable students to work as real engineers in an engineering community, in many cases the PBL practice lacks community and industry involvement, and students' output has few chances of being used in a real professional setting (Roach, Tilley, and Mitchell 2018; Ruhizan, Ramlee, and Azami 2009).

3.2.3. Culture level

The variety of cultural backgrounds: In PBL practice at the project level, especially in international PBL projects, students had more chances to collaborate with group members from other subjects or countries. With various cultural backgrounds, students had different ways of thinking, working habits, and paradigms of their subjects, which shaped their beliefs and world views and led to gaps in their understanding of others' perspectives, especially at the beginning of the teamwork process (Bani-Hani et al. 2018; O'Shea et al. 2013). Thus, students in these multicultural groups need to put more effort into their team-building process, improve their culture sensitivity, and learn to think from others' perspectives in the teamwork process.

Language barriers for non-native students: Language barriers were found to be another issue for participants in international PBL projects. In particular, for those non-native speakers in international project teams, low language proficiency become a barrier in teamwork processes (Guerra 2017; Khalaf et al. 2013; Ota and Punyabukkana 2016). Meanwhile, while working with students from other disciplines in these international PBL projects, non-native speakers without basic professional knowledge in specific fields faced more challenges because of their unfamiliarity with the terminology (Bani-Hani et al. 2018; Ota and Punyabukkana 2016).

4. Discussion

This paper scrutinises evidence regarding current PBL implementation and challenges based on empirical studies as a result of a search of four databases (EBSCO, ERIC, Web of Science, and SCOPUS) with relevant empirical studies. Based on the literature review, this research illustrates the diversity of PBL practice at all curriculum levels: course level, cross-course level, curriculum level, and project level. Across these four levels, similar challenges at the individual level for teachers and students, institutional level, and culture level were also reported in detail.

Although PBL methods were used in different ways among engineering education institutions, common patterns could be observed in their practice. No matter whether at course level, curriculum level, or project level, students formulated small working groups to solve open-ended, ill-structure, and real-life problems through self-directed learning, which follows the learning principles of PBL (Edström and Kolmos 2014). According to Savin-Baden's (2014) framework, the main types of current PBL practice found at all levels include PBL for practical capabilities and PBL for design-based learning, with learning objectives aimed at developing students' practical capabilities and design-based skills. In addition, similar assessment methods, such as team oral presentations and project written reports, were also widely used in current PBL practice at diverse levels (Costello 2017; Garcia-Robles et al. 2009; Ota and Punyabukkana 2016; Wallace and Webb 2017). Formative assessment has become an important part of the assessment system in PBL practices (Thomas 1997), and peer assessment, self-assessment, meeting records, and other process control tools were used to help both students and teachers to obtain a deeper understanding and reflection about students' learning outcomes in PBL (Palmer and Hall 2011; Qattawi, Venhovens, and Brooks 2014).

In spite of these common patterns, we can also see the huge variation in PBL implementation strategies among different levels. Based on Savin-Baden's (2014) framework of types of PBL methods, the main types of current PBL practice at the course level are PBL for knowledge management, project-led PBL, PBL for practical capabilities, and PBL for design-based learning, while more cases of PBL for critical understanding, PBL for multimodal reasoning, and collaborative distributed PBL were reported at the cross-course and curriculum level. At the PBL project level, in order to develop students' global horizon, the type of PBL transformation and social reform was used in an international PBL project. Specifically, student project groups showed different group diversity at various PBL practice levels. At course level, problem-oriented PBL was used with higher frequency, and students usually came from one subject and focused on problems in a specific field (Redkar 2012), while PBL at curriculum level and project level were more project-oriented and provided more opportunities for students to work in multidisciplinary teams or even to be able to make contact with clients from companies or industries (Tan and Shen 2018; Terrón-López et al. 2015). Especially in those international PBL projects, the capability of solving global issues became one of learning objectives, and students faced higher team diversity because they needed to work with people with various cultural backgrounds and subject backgrounds (Salán Ballesteros et al. 2017; Ota and Punyabukkana 2016). Meanwhile, assessment systems of the PBL implementation at different levels had disparate component parts. At course level, several traditional assessment methods, such as midterm exams, individual final exams, quizzes, lab tests, and attendance, were used more frequently (Qattawi, Venhovens, and Brooks

2014; Simcock et al. 2007; Tan and Shen 2018), while short-term or international PBL projects were more focused on students' project experiences as well as the global horizon (Salán Ballesteros et al. 2017; Ota and Punyabukkana 2016).

While PBL implementation varies from the course level to the project level, similar challenges were reported based on empirical research from both individual and system perspectives. From individual perspectives, not only students but also engineering staff are facing the challenge of transferring from traditional learning methods into PBL methods (Van Barneveld and Strobel 2019). Challenges in terms of heavy workload, limited time and resources, and a lack of PBL pedagogy training for both students and teachers were frequently mentioned in articles from course and cross-course levels to curriculum and project levels (Chan 2016; Clyne and Billiar 2016; Dos Santos et al. 2009; Du et al. 2019; Hassan et al. 2015). Meanwhile, no matter whether using PBL elements in a course or using a curriculum PBL strategy, supportive policy and resources in aspects of finance support, promotion standards, infrastructure and learning equipment from the institutional level are needed to smooth PBL practice processes (Arman 2019; Clyne and Billiar 2016; Du et al. 2019). Teachers call for building a communication bridge between units and departments in an institution, which could provide them with more chances to share experience, learn about PBL pedagogy, choose effective assessment tools, and improve their facilitation skills (Chan 2016; Kjellberg, Adawi, and Brolin 2015).

Nevertheless, PBL practice at one level might also face particular challenges because of specific PBL implementation strategies. Based on analysis of selected articles, PBL at the course level is the majority use in current PBL implementation, in accordance with Thomas's review work (1997). However, PBL within course might not be an optimal use of PBL since one significant challenge for PBL in a single course is the lack of repercussions in the curriculum, which limits the long-term effectiveness of PBL in terms of benefiting students (García-Martín and Pérez-Martínez 2017). Also pointed out by Strobel and Van Barneveld (2009), the duration of PBL could influence the effectiveness of PBL. Compared with PBL within one course or one semester, students in PBL at the cross-course and curriculum level might be superior when it comes to students' familiarity with PBL, transferable skills development and satisfaction of teamwork and facilitation. Thus, in order to improve students' longitudinal development through PBL, one suggestion is to design a structured PBL implementation plan, and to consider the mixed usage of PBL methods at different levels. Specific learning objectives and learning activities in PBL context should be set for students at different learning stages. Not only teachers from one single course who adopt PBL methods should consider students' expected learning outcomes, but also the whole faculty need collaboration to design a mixed mode with combination of PBL at different levels, aimed at achieving students' longitudinal development and benefits from PBL. In addition, with more international education collaboration, compared with previous review on PBL (Thomas 1997), recent year studies reported more challenges related culture and background differences (Beddoes, Jesiek, and Borrego 2010). In cross-institution or international PBL projects, participants might face more challenges at culture background level because participants from diverse subjects and countries have various ways of thinking (Bani-Hani et al. 2018). For non-native speakers, language barriers were reported as another kind of difficulty to overcome, which could influence their learning experience and outcomes in PBL (Khalaf et al. 2013).

With the call for exploring the variations of PBL implementation (Van Barneveld and Strobel 2019), this study provides an overview of current various PBL implementation and challenges over the past 20 years, which could inspire our reflection and design of future PBL practice. Although PBL elements or models were added to the courses or curriculum design across diverse higher education institutions, no matter at which level, we see challenges, faced by students, teachers, and engineering educators, that illustrate the difficulty of applying any variation of PBL design in the curriculum. The evidence from this study suggests that for engineering faculty and universities, a structured PBL design is needed. PBL training for students and academic staff, support from the institutional level, and coherent curriculum design are needed to optimise current PBL practices. For engineering

education researchers, this study summarises divers PBL implementation and provides a significant thought of combining PBL methods at different levels and assessing the effectiveness of those PBL design for students' sustainable development. More empirical evidence is needed to illustrate the efficiency of PBL at different levels, give instructions, and share experiences for better implementation of PBL strategies.

The main limitation of this research was that we only reviewed and analysed studies in four databases in the past two decades. There is a lack of further information about PBL practice strategies reported in non-English journals, institutional reports, non-academic articles, and other databases in engineering education, which may possibly lead to bias in our findings. However, the audit and peer-review processes could minimise the risk and identify the results more subjectively (Murzi 2014). Another limitation of this research is that it does not show many details about students' teamwork, learning experience, and learning outcomes in PBL practice. Future work will extend the current inclusive scope with more databases and selected durations in order to track the development and changes in PBL implementation over decades in engineering education. More details of learning objectives, PBL principles, students' learning outcomes, learning and teamwork experiences, and impact factors in current PBL practice will also be coded and analysed to provide guidance for engineering educators to optimise the current PBL course, curriculum, and programme design. The trends of PBL development in different counties and regions is also an interesting topic to explore and needs more attentions. Moreover, we also call for more empirical evidence of the effectiveness of PBL at different levels as well as PBL design with mixed mode/level. When various PBL practice are facing similar challenges, more exploration is needed through empirical research to share effective strategies and experience, cope with current challenges, and foresee the future development directions of PBL.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Table A1. PBL implementation at different levels.

Practice level	Problem for students to solve	Project duration	Group size	Assessment methods
Course level	$N = 73$ - Well-defined problems given by teachers, or proposed by students (Lobbestael and Sleep 2016; Gratchev and Jeng 2018; Selçuk et al. 2011) - In-class semi-structured problems (Dong and Guo 2014; Garcia-Robles et al. 2009) - Open-ended problems given by teachers (García-Martín and Pérez-Martínez 2017; Mantri et al. 2009; Warnock and Mohammadi-Aragh 2016; Wengrowicz, Dori, and Dori 2017; Zhang, Hansen, and Andersen 2016), defined by students (Balve, Krüger, and Tolstrup Sørensen 2017; Dong and Guo 2014; Gratchev and Jeng 2018; Kunberger 2013; Nasr and Ramadan 2008) or proposed by companies (Rad et al. 2017) - Real-world/life problems given by teachers (Finí, Parast, and Sarin 2011; Calvo et al. 2018; Garcia-Robles et al. 2009; Hugo, Brennan, and Gu 2012; Sahin 2010) or defined by students (Kim 2012; Sedaghat et al. 2017; Zhang, Xie, and Li 2018) - Ill-structured problems given by teachers (Hartman and Gindy 2010; Huang and Yang 2008) or defined by students (Al Mughrabi and Jaeger 2018; Pinho-Lopes 2018) - Interdisciplinary problems given by teachers (Hartman and Gindy 2010. etc.) or defined by students (Hassan et al. 2015) - Cases given by teachers (Aranzabal, Epelde, and Artetxe 2019; Chaparro-Peláez et al. 2013)	- Two weeks (Ahern 2010) - Three weeks (Williams 2011; Zhang and Liu 2012) - Four weeks (Bessa and Santos 2017) - Nine weeks (Gweon et al. 2017; Martinez-Rodrigo et al. 2017) - Ten weeks (Macias-Guarasa et al. 2006; Roach, Tilley, and Mitchell 2018) - One month (Hugo, Brennan, and Gu 2012) - Two months (Ribeiro and Mizukami 2005) - One semester (12–18 weeks) (Al Mughrabi and Jaeger 2018; Mellon and Ramlí 2017; Podges, Kommers, and van Joolingen 2014)	Small groups (3–8 students in one team)	- Team oral presentation marked by exports (Costello 2017; Garcia-Robles et al. 2009; Hassan et al. 2015; Pinho-Lopes 2018; Roth 2016) - Individual oral presentation (Clyne and Billiar 2016; Zhang and Liu 2012) - Written reports marked by exports (Costello 2017; Hassan et al. 2015; Kim 2012; Mantri et al. 2009; Pinho-Lopes 2018) - Poster marked by other groups (Sahin 2010) - In-group peer assessment (Chaparro-Peláez et al. 2013; Zou and Mickleborough 2015) - Out-group peer assessments (Chaparro-Peláez et al. 2013; Hugo, Brennan, and Gu 2012; Lutsenko 2018; McIntyre 2003; Santos-Martin et al. 2012) - Learner Agent Object (LAO) portfolio (Abu-Aisheh et al. 2016) - Homework assessment (Kim 2012; Nasr and Ramadan 2008; Rodríguez Montequín et al. 2013) - Self-assessment (Hu, Ortiz, and Sriraman 2014) - Lab reports (Clyne and Billiar 2016) - Observation (Nasr and Ramadan 2008; Russell et al. 2010; Gweon et al. 2017) - Midterm exams (Dong and Guo 2014; Gratchev and Jeng 2018; Mantri et al. 2009) - Traditional final exams (Chan 2016; Gratchev and Jeng 2018; Hassan et al.

(Continued)

Table A1. Continued.

	Practice level	Problem for students to solve	Project duration	Group size	Assessment methods
					2015; Lamar et al. 2012 ; Nasr and Ramadan 2008; Pinho-Lopes 2018) • Individual quizzes (Aranzabal, Epelde, and Artetxe 2019; García-Martín and Pérez-Martínez 2017) • Attendance (Costello 2017; Mantri et al. 2009)
Cross-course level	N = 6	• Real-life design or problems (Bani-Hani et al. 2018; Calvo et al. 2018; Chen and Teng 2011; Kay et al. 2000; MacCrum 2017) • Open-ended (Kay et al. 2000; MacCrum 2017) • Authentic (Kay et al. 2000) • Given by teachers (Calvo et al. 2018; Chen and Teng 2011; Kay et al. 2000) • Proposed by students (MacCrum 2017)	• 11 weeks (MacCrum 2017) • One semester (Bani-Hani et al. 2018; Calvo et al. 2018; Kay et al. 2000) • Two semesters (Chen and Teng 2011)	3–6 students in one team	• Group presentation (Calvo et al. 2018; Kay et al. 2000; MacCrum 2017) • Group written reports (Kay et al. 2000; MacCrum 2017) • Individual reflective reports (Kay et al. 2000) • Weekly plan (Kay et al. 2000) • Peer assessment (Chowdhury 2016) • Meeting records (Chen and Teng 2011) • Midterm exam (Calvo et al. 2018) • Final individual exam (Calvo et al. 2018)
Curriculum level	N = 23	• Real-world problems proposed by students (Quinn and Albano 2008) • Open-ended problems (Van Barneveld and Strobel 2011; Hernandez, Ravn, and Valero 2015; Hung 2011; O'Shea et al. 2013; Tan and Shen 2018) • Ill-structured problems (Van Barneveld and Strobel 2011) • Interdisciplinary problems (Dahms, Spliid, and Nielsen 2017; Qattawi, Venhovens, and Brooks 2014; Tan and Shen 2018) • Product-oriented design (Li 2015; Schultz 2018) • Case study (Hosseinzadeh and Hesamzadeh 2012) • Defined by students (Ahern 2010; Van Barneveld and Strobel 2019; Dobson and Tomkinson 2012; O'Shea et al. 2013; Servant and Dewar 2015) • Defined by teachers (Hosseinzadeh and Hesamzadeh 2012; Li 2015; Palmer and Hall 2011; Schultz 2018) • Defined by students or teachers together in different steps (Christensen 2008; Dahms, Spliid, and Nielsen 2017) • Defined by clients or companies (Terrón-López et al. 2015; Wallace and Webb 2017; Zhou 2012)	• Use PBL in the first two years in undergraduate projects (Van Barneveld and Strobel 2019; Servant and Dewar 2015) • Use PBL in the first three years (Perrenet, Bouhuijs, and Smits 2000) • Use PBL in the last two years (Ahern 2010) • Use PBL in four years (Hosseinzadeh and Hesamzadeh 2012; Simcock et al. 2007) • Report a one-year PBL example (Balve, Krüger, and Tolstrup Sørensen 2017; Palmer and Hall 2011; Schultz 2018; Terrón-López et al. 2015) • Report a one-semester PBL example (Dahms, Spliid, and Nielsen 2017; Tan and Shen 2018; Wallace and Webb 2017) • Report a two-week PBL example (Li 2015)	• 2–3 students (Li 2015; Tan and Shen 2018; Zhou, Nielsen, and Kolmos 2011) • 4–5 students (Terrón-López et al. 2015; Hosseinzadeh and Hesamzadeh 2012) • 6–8 students (Ahern 2010; Christensen 2008; Dahms, Spliid, and Nielsen 2017; O'Shea et al. 2013)	• Team oral presentation marked by exports (Wallace & Webb; Ahern; Dahms, Spliid, and Nielsen 2017; Hung 2011; Li 2015; Tan and Shen 2018) • Team written reports marked by exports (Ahern 2010; Dahms, Spliid, and Nielsen 2017; Li 2015; Tan and Shen 2018; Wallace and Webb 2017) • Posters marked by other groups (Ahern 2010) • Weekly group relationship assessment (O'Shea et al. 2013) • Peer evaluation (Palmer and Hall 2011; Qattawi, Venhovens, and Brooks 2014) • Individual essay (Tan and Shen 2018) • Individual project report (O'Shea et al. 2013) • Exams (Qattawi, Venhovens, and Brooks 2014; Simcock et al. 2007; Tan and Shen 2018)



Project level	N = 6	• Real-world problems or design (Du, Su, and Liu 2013; Mantelet, Segonds, and Jean 2018; Mativo et al. 2018; Moliner et al. 2019; Sanger and Ziyatdinova 2015) • Open-ended problems (Ota and Punyabukkana 2016) • Complex (Sanger and Ziyatdinova 2015) • Practical problems (Ota and Punyabukkana 2016) • Multidisciplinary (Du, Su, and Liu 2013) • Given by companies (Sanger and Ziyatdinova 2015) • Identified by students (Du, Su, and Liu 2013) • Decided by students and teachers together (Moliner et al. 2019; Ota and Punyabukkana 2016)	• One small project a day (Mantelet, Segonds, and Jean 2018) • Four-week project (Mativo et al. 2018) • One project a year, two years in total (Sanger and Ziyatdinova 2015) • 15 weeks (Ota and Punyabukkana 2016) • Six months (Du, Su, and Liu 2013)	• 2–4 students (Ota and Punyabukkana 2016; Mativo et al. 2018) • 5 students (Moliner et al. 2019) • 7–9 students (Du, Su, and Liu 2013; Mantelet, Segonds, and Jean 2018)	• Attendance or participation level (Qattawi, Venhovens, and Brooks 2014; Tan and Shen 2018; Terrón-López et al. 2015) • Individual and team video (Moliner et al. 2019) • Project written reports (Du, Su, and Liu 2013; Mativo et al. 2018; Moliner et al. 2019; Ota and Punyabukkana 2016) • Team presentation (Du, Su, and Liu 2013; Mantelet, Segonds, and Jean 2018; Ota and Punyabukkana 2016) • Meeting/collaboration reports (Du, Su, and Liu 2013; Moliner et al. 2019; Ota and Punyabukkana 2016) • Process assessment by experts (Mantelet, Segonds, and Jean 2018) • Frequency of platform usage (Moliner et al. 2019) • Individual essay (Ota and Punyabukkana 2016)
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Tips: All codes based on description of PBL practice are shown in this table, but not all articles are listed as reference examples. In those articles, descriptions of problem types, project duration, and assessment methods are familiar with other practices, so they are omitted because of the limited space.

Table A2. Challenges reported in selected articles.

	For teachers	For students
Individual level	Lack of training as facilitators • How to facilitate students' teamwork (Bani-Hani et al. 2018) • Deal with issues in group division processes (Bani-Hani et al. 2018) • The transfer from lecturers to facilitators (Chan 2016; Clyne and Billiar 2016; Lutsenko 2018) • Lack of teaching skills in active/contextual learning (Clyne and Billiar 2016) • Unfamiliarity with PBL pedagogy (Santos-Martin et al. 2012) • Need practical experience in one or multidisciplinary fields (Nasr and Ramadan 2008) • How to ensure the effectiveness of PBL, such as team size; students' participation (Hugo, Brennan, and Gu 2012; Dos Santos et al. 2009; Ribeiro and Mizukami 2005) • How to balance between helping with and influencing the students' work (Marti et al. 2015; Zhou, Nielsen, and Kolmos 2011) • Deal with individual differences of students (Zhang, Xie, and Li 2018) Challenges of assessment • How to assess effective teamwork (Bani-Hani et al. 2018; Clyne and Billiar 2016; Gweon et al. 2017) • Difficulties in assessing transferable skills and actual learning (Chan 2016; Chaparro-Peláez et al. 2013; Kunberger 2013; Macias-Guarasa et al. 2006; Marti et al. 2015) • How to make students remain honest in peer assessment (Ahern 2010) • The choice of assessment tools (Dos Santos et al. 2009) • How to establish rules to evaluate team results and individual learning outcomes (Lutsenko 2018; Pinho-Lopes 2018; Ribeiro and Mizukami 2004) Set difficulty levels of problems for students • How to ensure the difficulty level of the problems for students (Lobbestael and Sleep 2016; Zhang, Xie, and Li 2018; Mokhtar 2011) • How to balance long-term knowledge retention vs immediate learning outcomes (Van Barneveld and Strobel 2019) Interpersonal challenges • Feel uncomfortable with unknown things (Wallace and Webb 2017) • Feeling of anxiety (Servant and Dewar 2015; Zhou 2012)	Lack of teamwork skills • How to communicate and interact with team members (Chaparro-Peláez et al. 2013; Kjellberg, Adawi, and Brolin 2015) • Difficulties for new members in integrating into friend-knowing teams (Bani-Hani et al. 2018) • Task division (Bani-Hani et al. 2018) • How to deal with those who don't work (Bani-Hani et al. 2018) • Collaborate with students with diverse backgrounds (Dong and Guo 2014; Pinho-Lopes 2018) • Deal with disagreement or conflict (McQuade et al. 2018; Ribeiro and Mizukami 2004; Perrenet, Bouhuijs, and Smits 2000) • Effective communication with supervisors (Balve, Krüger, and Tolstrup Sørensen 2017) • Lack of leadership training for students (Kjellberg, Adawi, and Brolin 2015) • Gender issues (Bani-Hani et al. 2018) Lack of self-learning skills • How to identify and deal with the messiness of the problems (Ahern 2010; Hu, Ortiz, and Sriraman 2014; Valdez et al. 2009) • How to translate knowledge into practical problem solutions (Bledsoe and Flick 2012; Lutsenko 2018) • Transfer from accepting knowledge or solutions provided by professors (Clyne and Billiar 2016) • Lack of learning motivation (Gratchev and Jeng 2018; Ribeiro and Mizukami 2004) • Lack of individual reflections (Williams 2011) • Different learning outcomes among different team members (Mehta 2002) • Meet technical difficulties and issues (Sedaghat et al. 2017) Lack of project management skills • How to make time schedule (Bani-Hani et al. 2018; Mehta 2002) • Need more time and effort (Clyne and Billiar 2016; Lamar et al. 2012; Martinez-Rodrigo et al. 2017; Palmer, 2011; Zhang, Xie, and Li 2018) • How to define project work steps (Lutsenko 2018) • Lack project management tools • Insufficient time for out-of-class learning and social activities in PBL projects (Sahin 2010)



Prior experience

- Pre-college learning culture and thinking styles (Chan 2016)
- Lack previous knowledge (O'Shea et al. 2013; Santos-Martin et al. 2012; Shen et al. 2012)

Interpersonal challenges

- Lack of self-confidence (Shen et al. 2012)
- Feeling of depression and anxiety in PBL (Kay et al. 2000; Zhou 2012)
- One's belief and ownership, such as doubting the effectiveness of PBL (Terrón-López et al. 2017; Quinn and Albano 2008)

Institutional
level

Time and effort devotion

- Heavy workload to prepare and guide PBL projects (García-Martín and Pérez-Martínez 2017; Hassan et al. 2015; Pinho-Lopes 2018; Zhang, Xie, and Li 2018)
- Need to devote much more time (Dos Santos et al. 2009; Lamar et al. 2012)
- The grade-oriented culture among engineering students (Williams 2011)

Lack of support from department or institution level

- Lack of communication, training, and support from colleagues in the department and faculty (Chan 2016; Kjellberg, Adawi, and Brolin 2015)
- Lack of materials and resources from institutions to support PBL (Clyne and Billiar 2016; Hosseinzadeh and Hesamzadeh 2012)
- Lack of supportive policy from universities (Arman 2019)
- Lack of involvement and collaboration of different units within a university (Arman 2019)
- Insufficient facilitators to guide and help students in the curriculum (MacCrum 2017; Sahin 2010; Setiawan 2019)

The way in which PBL is applied

- Learning principles (Dos Santos et al. 2009)
- PBL models (Dos Santos et al. 2009)
- Balance between courses and projects (Kim 2012)
- PBL in single courses lack great repercussions in the curriculum (Calvo et al. 2018; García-Martín and Pérez-Martínez 2017)

Limitation of objective conditions

- Difficulty in controlling learning process in large class (Clyne and Billiar 2016; Huang and Yang 2008; Ruhizan, Ramlee, and Azami 2009)
- Lack of technical support (Bani-Hani et al. 2018)
- Lack of community and industry involvement (Ruhizan, Ramlee, and Azami 2009)
- Students' outputs are not used in a real professional setting (Li 2015; Roach, Tilley, and Mitchell 2018)
- Need more space and freedom for students to explore and determine their learning (Ribeiro and Mizukami 2005)

Cultural level

Nationality and culture sensitivity (Bani-Hani et al. 2018)
 Different cultures and different ways of thinking (Bani-Hani et al. 2018; Guerra 2017; O'Shea et al. 2013)
 Language barriers for non-native students (Khalaf et al. 2013; Ota and Puniyabukkana 2016)